Analysis of the CDC Alzheimer's Disease and Healthy Aging Data

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Dataset: CDC Alzheimer's and Healthy Aging Data

Dataset ID: hfr9-rurv

Introduction

The purpose of this analysis was to look at patterns in reported health outcomes among older adults using the CDC Alzheimer's Disease and Healthy Aging Data. The analysis primarily focused on subjective cognitive decline (SCD), reported lifetime depression, and differences between adults aged 50-64 and adults aged 65 years or older. The project investigated two main questions: whether reported SCD prevalence differs between the two age groups, and whether the association between reported lifetime depression prevalence and SCD differs by age group. Additional comparisons examined obesity and current-smoking prevalence across the same age groups. Outlier analyses were also conducted to evaluate how extreme estimates affected the results. Because the CDC dataset contains aggregated survey estimates rather than individual-level medical records, the findings describe patterns in reported prevalence. They should not be interpreted as evidence of individual diagnoses or causal relationships.

Dataset Description

The CDC Alzheimer's Disease and Healthy Aging Data contains 284,142 observations and 33 variables covering 2015–2022. It includes 59 locations, seven health classes, and 39 health topics. The data source is the Behavioral Risk Factor Surveillance System (BRFSS).

The dataset includes variables describing the year, location, health class, topic, question, reported estimate, age group, and demographic stratification. The primary numerical field used in the analyses was data_value, which contains the reported health estimate. The dataset contains 22,182 observations in the Cognitive Decline class and 5,854 observations associated specifically with the subjective cognitive decline or memory loss topic. Data completeness was an important consideration. Although the overall missing-data rate across all dataset cells was 9.53%, the data_value variable had 91,334 missing observations, representing 32.14% of its values. Confidence-limit fields also had approximately 32.22% missingness. These missing values were considered during filtering and analysis. The dataset reports both percentage-based and mean-based health measures. Therefore, values from different topics were not treated as directly comparable when they represented different measurement types.

Differences in SCD by Age

The first major analysis examined whether reported SCD prevalence differed between adults aged 50–64 and adults aged 65 years or older. These age groups were selected because they are the two age-specific categories directly provided by the CDC dataset. The analysis included 1,286 observations for the 50–64 group and 1,260 observations for the 65+ group.

Statistic	Ages 50–64	Ages 65+
Observations	1,286	1,260
Mean prevalence	11.14%	12.20%
Median prevalence	10.70%	11.80%
Standard deviation	3.99	4.21

The 65+ group had a higher mean reported SCD prevalence than the 50–64 group, with a difference of approximately 1.06 percentage points. The median was also higher among adults aged 65 years or older. An independent two-sample t-test produced a t-statistic of 17.52 and a p-value of approximately 1.42×10^{-67}. At a significance level of 0.05, the null hypothesis was rejected. The results indicate a statistically significant difference in reported SCD prevalence between the two age groups.

Outlier Analysis

Outliers were identified using the 1.5 × IQR rule. The analysis identified 54 outliers (4.20%) in the 50–64 group and 69 outliers (5.48%) in the 65+ group. After removing these observations, the mean prevalence decreased from 11.14% to 10.68% for ages 50–64 and from 12.20% to 11.86% for ages 65+. The medians changed only slightly. This suggests that extreme estimates had a greater effect on the means than on the medians. A majority of the identified outliers had associated CDC footnotes. This is important because some extreme estimates may reflect reporting limitations or regional-estimate warnings rather than incorrect data. Therefore, IQR-defined outliers were not automatically treated as erroneous observations.

4. Association Between Depression and Subjective Cognitive Decline

The primary relationship analysis examined whether reported lifetime depression prevalence was associated with reported SCD prevalence, and whether this association differed between the two age groups. Depression and SCD estimates were matched by year, location, and age group. This produced 253 matched observations for adults aged 50–64 and 251 matched observations for adults aged 65 years or older. For adults aged 50–64, the Pearson correlation was 0.441, indicating a moderate positive association between reported lifetime depression prevalence and reported SCD prevalence. In other words, matched estimates with higher reported depression prevalence tended to have higher reported SCD prevalence. For adults aged 65 years or older, the Pearson correlation was 0.177, indicating a weaker positive association. This association was statistically significant, with a p-value of 0.00492. The results suggest that the relationship between reported depression and SCD was stronger among adults aged 50–64 than among adults aged 65 years or older. However, correlation does not establish that depression causes SCD or

that SCD causes depression. Both measures are aggregated survey estimates, and other factors may contribute to the observed relationship.

Regression Diagnostics

Linear regression models were examined separately for the two age groups. Residual plots were generally centered around zero, with no obvious systematic curved pattern. The plots did not show strong visual evidence of substantial heteroscedasticity. The Q-Q plots showed that residuals followed the reference line reasonably closely through much of the distribution, particularly around the center. However, deviations were visible in the lower and upper tails, indicating that the residuals were not perfectly normally distributed. Overall, the diagnostic plots provided visual support for using a linear model to describe the associations, while also indicating that extreme residuals and departures from normality should be considered when interpreting the results.

5. Additional Health-Measure Comparisons

Additional analyses compared reported obesity and current-smoking prevalence between the two age groups.

Obesity

The mean reported obesity prevalence was 36.696% among adults aged 50–64 and 28.988% among adults aged 65 years or older. The difference was approximately 7.708 percentage points. After IQR-based outlier removal, the means were 36.664% and 29.245%, respectively. The difference decreased to approximately 7.420 percentage points. The Welch two-sample t-test remained statistically significant after outlier removal.

Current Smoking

The mean reported current-smoking prevalence was 18.348% among adults aged 50–64 and 9.130% among adults aged 65 years or older. The difference was approximately 9.218 percentage points. After outlier removal, the means changed only slightly, to 18.317% and 9.112%. The difference remained approximately 9.205 percentage points, and the statistical test remained significant. These comparisons show that reported obesity and current-smoking prevalence differed between the age groups in this dataset. The small changes after outlier removal suggest that the observed differences were not driven primarily by the identified extreme estimates.

6. Cross-Feature Outlier Analysis

A further analysis examined whether outliers occurred in the same matched samples across three health measures: current smoking, lifetime diagnosis of depression, and SCD. Among 923 matched samples, 880 (95.34%) were not identified as outliers for any of the three features. The

remaining 43 samples (4.66%) were identified as outliers for exactly one feature. No samples were identified as outliers for two or more features. This finding suggests that extreme estimates generally occurred in different samples across the three measures. Consequently, removing an entire matched sample because it was an outlier for one feature could unnecessarily remove valid observations for another feature. Outlier handling should therefore be conducted separately for each health measure.

7. Limitations

Several limitations should be considered when interpreting these results. First, the CDC dataset contains aggregated survey estimates rather than individual-level observations. The analyses therefore describe relationships among reported estimates and cannot establish individual-level relationships or causal effects. Second, missingness in data_value affected 32.14% of the records. Analyses were consequently based on available non-missing estimates, which may affect the representativeness of the analyzed observations. Third, CDC footnotes identify reporting limitations for some estimates. These notes are especially relevant when interpreting extreme values and statistical outliers. Finally, the analyses compare age groups using the categories supplied by the dataset. The available data do not provide individual ages or narrower age categories, so more specific age comparisons could not be performed.

8. Conclusion

The analysis identified several differences in reported health estimates between adults aged 50–64 and adults aged 65 years or older. The 65+ group had higher mean and median SCD prevalence, while the 50–64 group had higher reported lifetime depression, obesity, and current-smoking prevalence in the comparisons conducted. The relationship between depression and SCD was positive in both age groups, but it was stronger among adults aged 50–64. Regression diagnostics provided visual support for a linear representation of these associations, although deviations in the residual tails should be considered. Outlier analyses showed that removing extreme estimates generally produced relatively small changes in the reported means. The cross-feature analysis also demonstrated that outliers did not usually occur in the same matched samples across all three health measures. Overall, the findings provide a descriptive and statistical examination of several health patterns in the CDC Alzheimer's Disease and Healthy Aging Data. Further work could investigate additional demographic and geographic differences, examine changes over time, and evaluate the robustness of the statistical results using methods that account for the structure of repeated, location-based survey estimates.